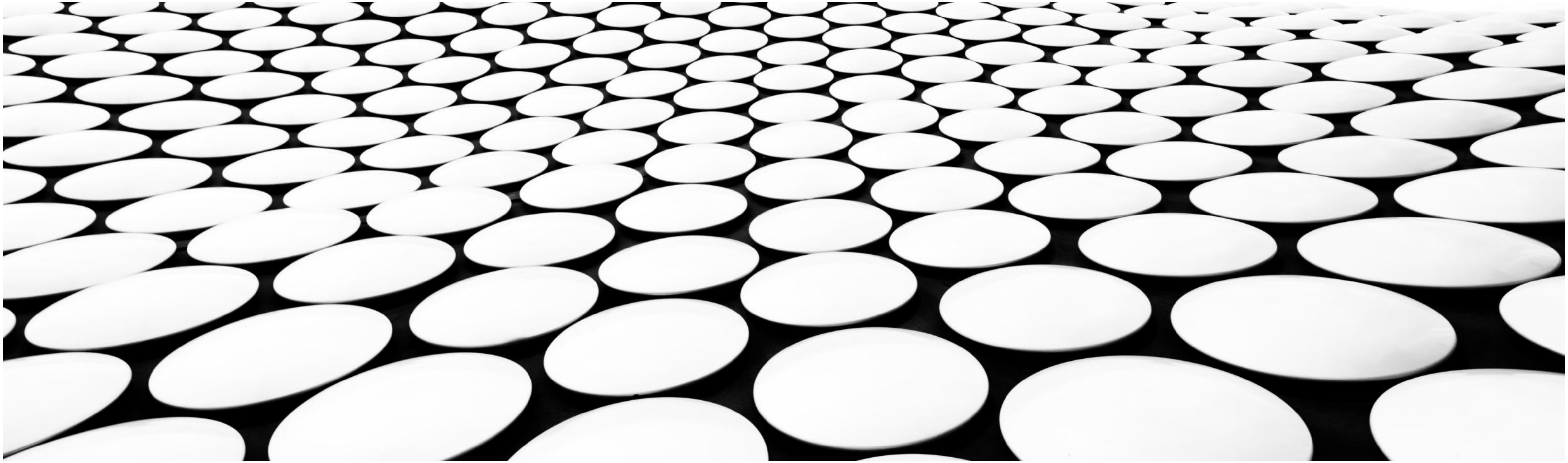
REVIEW SESSION



REVIEW SESSION

- Go through 30 example questions for the different weeks
- Point to important material throughout these questions
- Point to question styles throughout these questions

Missing Data

Slide deck: Week6-Imputation

Question 1

An interview questionnaire ask job applicants their age. A job applicant does not answer because he thinks he could be discriminated by being too old. Which type of missing value is that?

a) Missing Completely at Random (MCAR)

b) Missing at Random (MAR)

c) Missing Not at Random (MNAR)

d) None of the above

Some patients forgot to record their weight during a checkup, and their missing weight data is not related to other factors like age or health conditions.

MCAR

MAR

Patients who are older are less likely to report their smoking habits. The missingness of smoking data can be explained by the age of the patient, which is available.

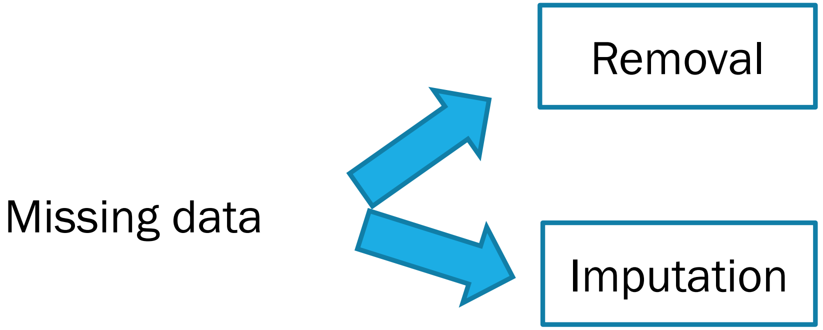
Patients with severe health conditions may avoid answering questions about their disease because they feel uncomfortable or stigmatized. The missingness is related to the severity of the condition.

MNAR

Question 2

A listwise removal algorithm will (1) select the rows containing more than one missing values and (2) remove the selected rows.

- True
- False



1) Listwise removal
2) Pairwise removal
3) Attribute removal

Question 3

Which type of imputation is examplified in the following scenario?

A survey asks people's weekly spending on food. For each missing value, the data scientist imputes it with a value randomly selected among the people who answered the question.

- a) Arbitrary value imputation
- b) Random Sample imputation
- c) Regression imputation
- d) Similarity based imputation

Fix

Missing data

- Removal
- Imputation

1) Arbitrary value imputation
2) Default value imputation
3) Random Sample imputation
4) Time-based imputation

univariate

5) Conditional imputation
6) Correlational imputation

bivariate

7) Regression imputation
8) Predictive imputation
9) Similarity-based imputation

multivariate

Question 4

Assuming the attribute with missing values is A and the correlated attribute is B. The linear correlation formula makes use of the mean value of A and the mean value of B.

- True
- False

Use the correlation between two numerical features to impute missing values.

Example:
If a feature *DistanceFromWork* is correlated with a feature T*ransportTimeToWork*, use their correlation to estimate missing values of T*ransportTimeToWork.*

- A common approach is using **linear correlation formulas:**

$$Y_{\text{missing}} = \rho \times \frac{\sigma_Y}{\sigma_X} \times (X - \mu_X) + \mu_Y$$

where:

- ρ is the Pearson correlation coefficient.
- μ and σ are the mean and standard deviation of X and Y.

Biases

Slide deck: Week6-Biases

Question 5

Which of the 3 selection bias is illustrated in the following scenario?

A food store wants to do a survey on lettuce. Cashiers are asked to give the survey to customers buying lettuce.

- a) Sampling bias
- b) Convergence bias
- c) Participation bias
- d) None of the above

Selection bias happens when the data set used for training is not representative enough, not large enough or too incomplete to sufficiently train a predictive system or obtain valid descriptive statistics.

Types of selection bias –

1. **Sampling bias:** occurs when randomization is not properly achieved during data collection.

2. **Convergence bias:** occurs when data is not selected in a representative manner.

3. **Participation bias:** occurs when the data is unrepresentative due to participations gaps in the data collection process.

Question 6

This type of bias can be insidious as different datasets would contain information referring to the same attribute (e.g. city population), but the gathering of that data would be performed differently in different cities.

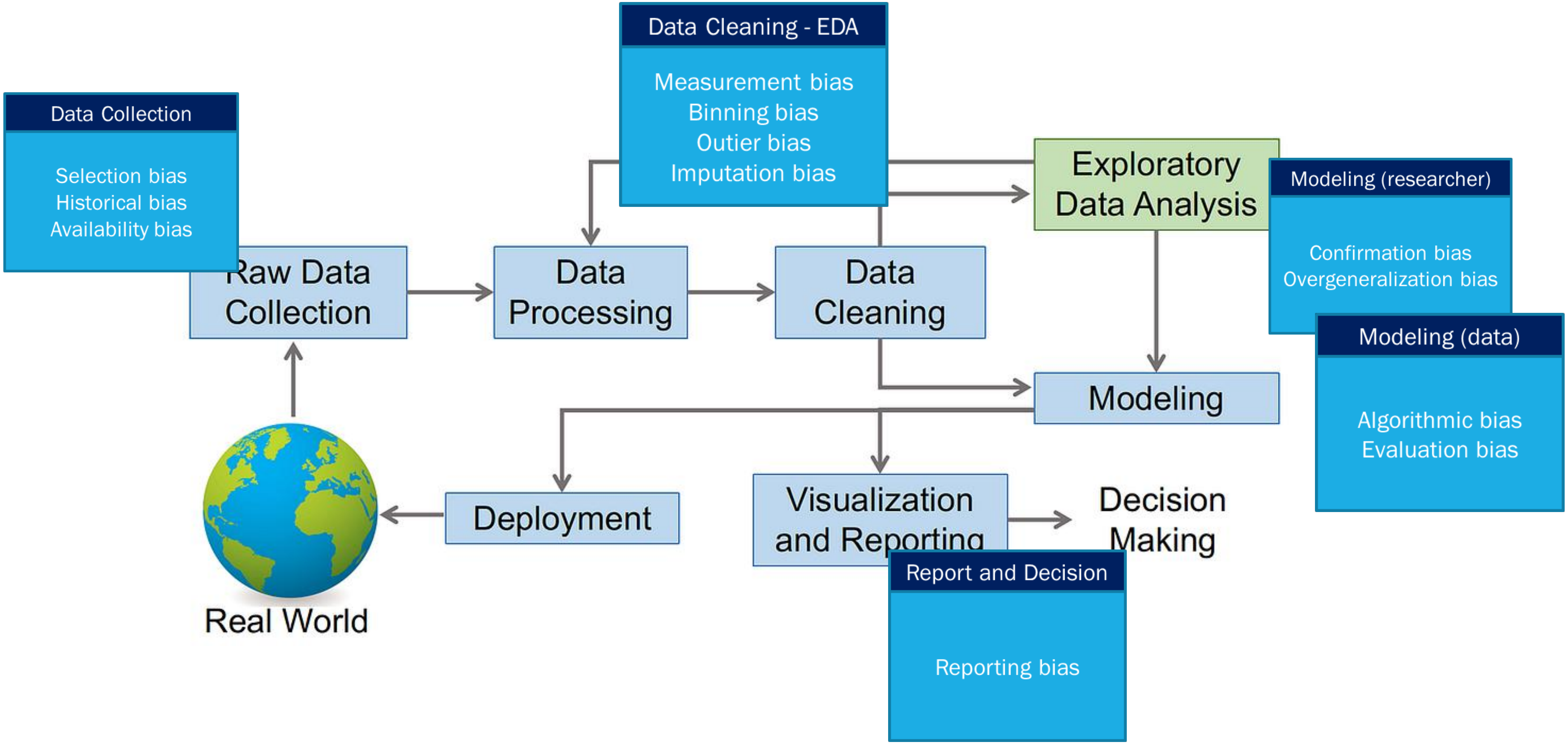
- a) Selection bias
- d) Measurement bias
- f) Outlier bias
- h) Algorithmic bias
- i) Evaluation bias
- l) Reporting bias

Question 7

Which of the biases is illustrated by the following example?

A predictive model is built using manager hiring data gathered in year 2000 containing many more male than female managers.

- a) Selection bias
- b) Historical bias
- c) Measurement bias
- d) Evaluation bias
- e) Overgeneralization bias
- f) Reporting bias


Data Collection
Selection bias
Historical bias
Availability bias
Data Cleaning - EDA
Measurement bias
Binning bias
Outier bias
Imputation bias
Raw Data Collection
Data Processing
Data Cleaning
Exploratory Data Analysis
Modeling (researcher)
Confirmation bias
Overgeneralization bias
Modeling (data)
Algorithmic bias
Evaluation bias
Modeling
Real World
Deployment
Visualization and Reporting
Decision Making
Report and Decision
Reporting bias

Supervised Learning

Slide deck: Week7-SupervisedLearning

Question 8

Indicate which statement(s) are true with respect to ***Supervised Machine Learning.***

- [x] Supervised Machine Learning requires labeled training data.
- [x] In Supervised Machine Learning, the goal is to train a model that will then be used for prediction on test data.
- [ ] In Supervised Machine Learning, the model built during training should be different than the model used during prediction.

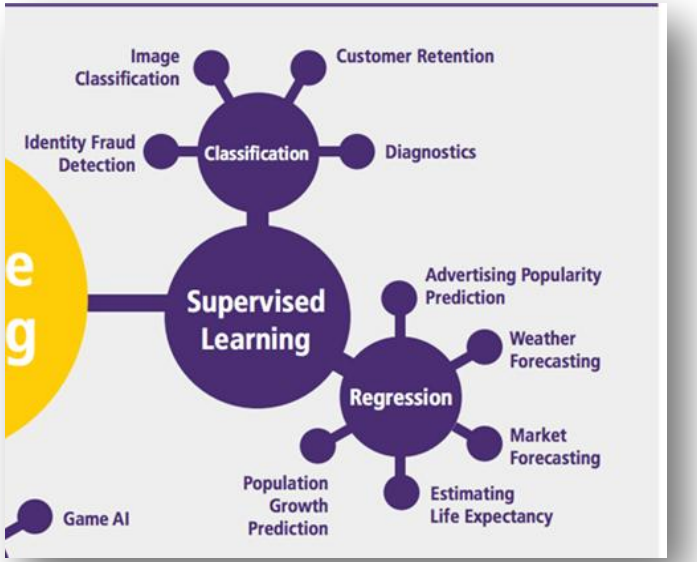
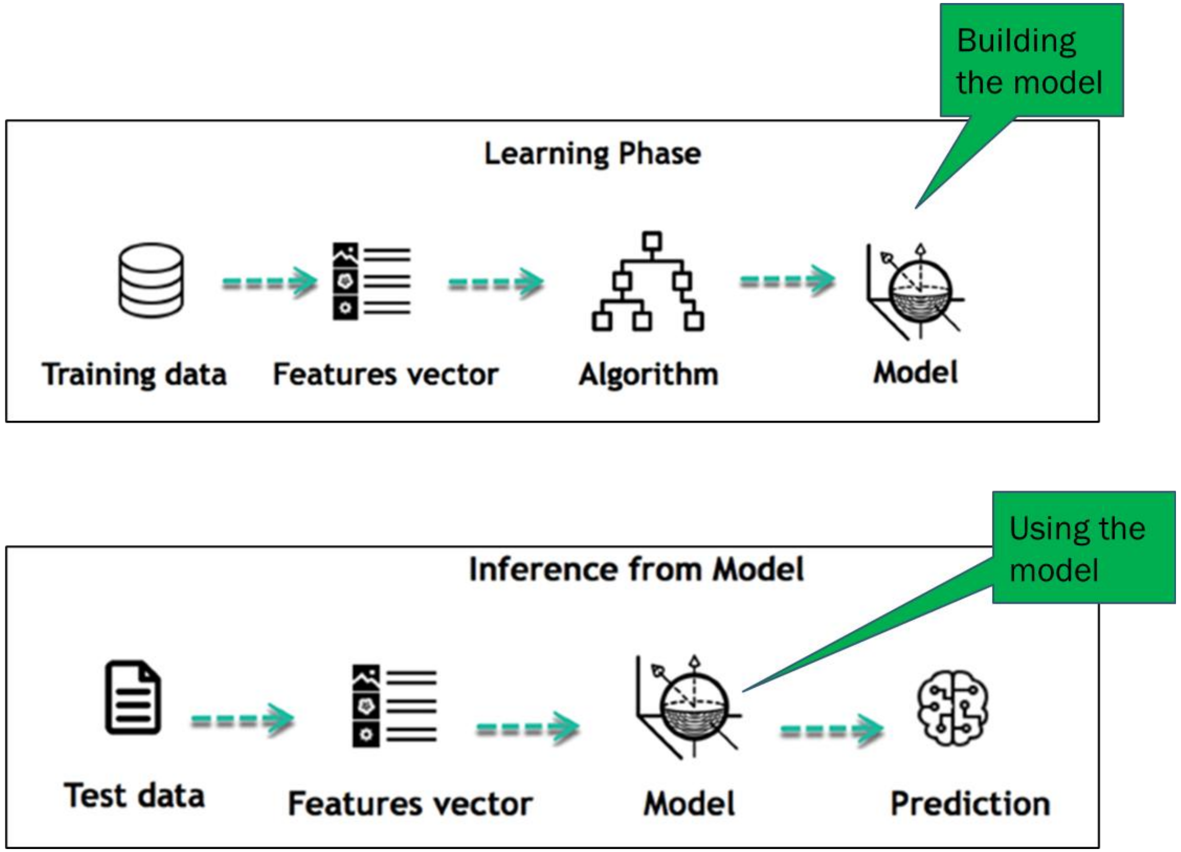


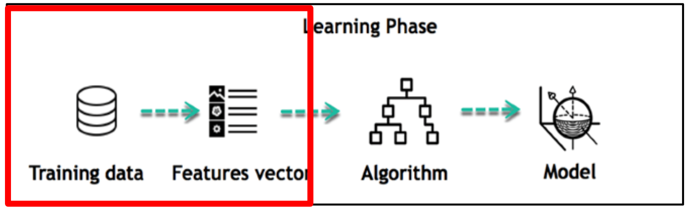
Image Classification
Customer Retention
Identity Fraud Detection
Classification
Diagnostics
Supervised Learning
Advertising Popularity Prediction
Weather Forecasting
Regression
Market Forecasting
Population Growth Prediction
Estimating Life Expectancy
Game AI


Building the model
Learning Phase
Training data
Features vector
Algorithm
Model
Using the model
Inference from Model
Test data
Features vector
Model
Prediction

Question 9

To encode numerical features into categorical features, we can use binning.

- True
- False



Representation of the data

	Features			Possible output	
Movie	Duration	Producer	Lead actor/actress	<u>Classification</u> Genre	<u>Regression</u> Gross Revenue
M1	90	P1	A1	Drama	3.2M
M2	140	P3	A2	Drama	200K
M3	80	P1	A1	Comedy	15M
M4	90	P1	A3	Drama	500K
M5	100	P3	A4	Drama	10K
M6	110	P2	A5	Comedy	1.3M
M7	100	P2	A5	Comedy	870K

Naive Bayes

Slide deck: Week7-NaiveBayes

Question 10

Assume an activity classification problem. The Table below provides the training set in which there are 3 attributes (rain, temperature, tiredness) and one output (activity).

Instance	Raining?	Temperature	Tired?	Activity
S1	Yes	Hot	No	Golf
S2	Yes	Warm	Yes	Bike
S3	Yes	Cold	No	TV
S4	No	Hot	No	Golf
S5	No	Warm	No	Golf
S6	Yes	Cold	No	TV
S7	Yes	Hot	Yes	TV
S8	No	Hot	Yes	TV
S9	No	Hot	No	Golf
S10	Yes	Cold	No	Bike

Evaluate P(Raining=Yes|Golf)

- 1/6
- 4/10
- 1/4
- 4/6

Suppose that the event E is (T=Average)

$$
\begin{aligned}
P(Bike|(Average) \quad &\propto P(Average|Bike) * P(Bike)\\
&\propto 2/9 * 9/15\\
&\propto 0.13
\end{aligned}
$$

Max

$$
\begin{aligned}
P(Drive|(Average) \quad &\propto P(Average|Drive)\ P(Drive)\\
&\propto 1/6 * 6/15\\
&\propto 0.06
\end{aligned}
$$

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

Question 11

Assume the Activity Classification system provides the following results, which we can compare to the Gold Standard for evaluation.

The system's precision for the class Golf is:

- 0.25
- 0.33
- 0.75
- 0.5

Test	Activity (Gold Standard)	Activity Prediction
S1	Golf	Bike
S2	Bike	Bike
S3	TV	TV
S4	Golf	Bike
S5	Golf	Bike
S6	TV	TV
S7	TV	TV
S8	TV	Golf
S9	Golf	Golf
S10	Bike	Golf

Precision = Tp / (Tp + Fp)
= 3 / (3 + 1) = 0.75

Recall = Tp / (Tp + Fn)
= 3 / (3 + 3) = 0.5

		Predicted		
		Bike	Not Bike	
Gold Standard	Bike	Tp = 3	Fn = 3	6
	Not bike	Fp = 1	Tn = 4	5
		4	7	11

Do the same precision/recall evaluation on the Drive class.

Linear Regression

Slide deck: Week7-LinearRegression

Question 12

Assuming we build a univariate linear regression model in which Blood Pressure is solely predicted with Weight. Assuming the slope of the model is 1 and its intercept is -10. What would be the Blood Pressure of Case 4.

- a) 87
- b) 89
- c) 95
- d) 93

Assuming the following small dataset.

	metric	metric	metric	metric
Case	**Blood Pressure**	**Age**	**Weight**	**Cholesterol**
1	78	58	76	153
2	91	48	70	234
3	79	34	52	195
4		62	99	190

$-10 + 99 = 89$

DATAtab

Source

Medical example

	metric	metric	metric	metric
Case	**Blood Pressure**	**Age**	**Weight**	**Cholesterol**
1	78	58	76	153
2	91	48	70	234
3	79	34	52	195
4	93	62	99	190
5	85	27	62	243
6	90	40	55	223
7	92	58	81	183
8	73	38	69	207
9	93	42	99	202
10	86	30	73	246
11	92	30	93	234
12	95	43	84	225
13	86	55	73	204

Multiple Linear Regression

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Question 13

What are possible factors for stopping the gradient descent algorithm?

- [ ] a certain preset number of iterations is done
- [ ] the predicted values are close enough (under a set error threshold) to the target values

Learning an LR Model with Gradient Descent

Learning the parameters can be done in an iterative manner, with an algorithm such as gradient descent

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + b_0$$

Initialize parameters at random

Repeat:

For each example in training set:

a. Predict $\hat{y}$ (forward pass)

b. For each parameter b_i

   a. Calculate derivative of error $\boxed{\frac{dE}{db_i}}$

   b. Update parameter

$$b_i^{(t)} = b_i^{(t-1)} - \alpha \frac{dE}{db_i}$$

Until "done"

What is « done »?

certain number of iterations

Convergence

Decision Trees

Slide deck: Week7-DecisionTrees

Question 14

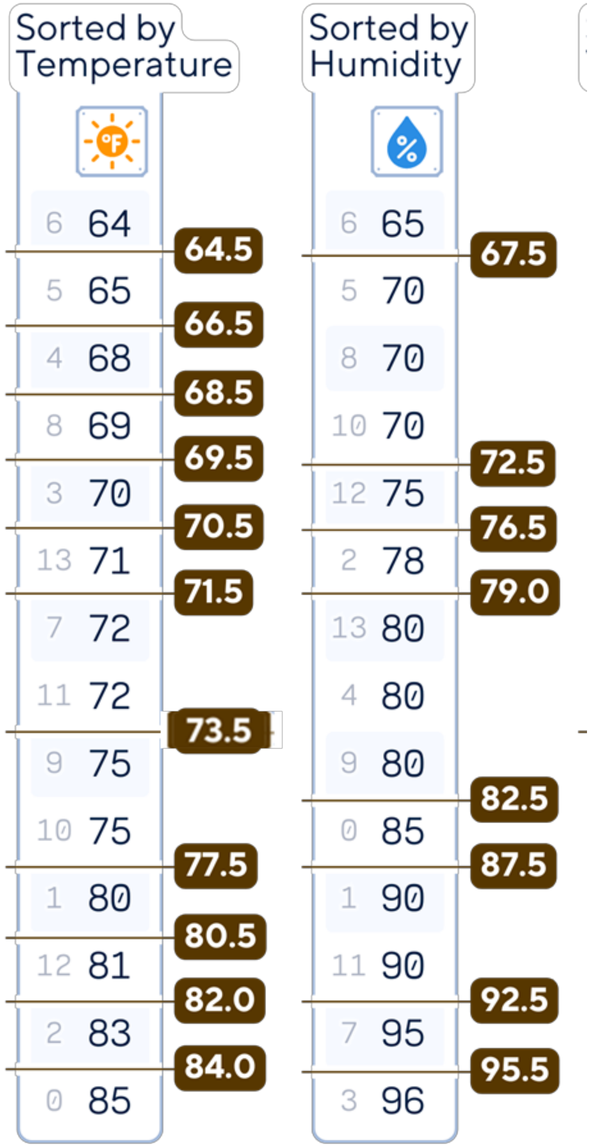
Provided 2 numerical attributes, on which we wish to build a decision tree, what is true based on the following image:

- a) A candidate splitting point corresponds to the median between two consecutive values.
- b) If for humidity, the values following 85 had been 95, 95, 95, 96, then we would have one additional splitting point than what is shown on the figure.

2. For each feature:

   a. Sort the feature values.
   b. Consider all possible thresholds between adjacent values as potential split points.

There will be lots of split points to check…



Question 15

Provided the following dataset, where we see the target (playing golf or not) and we see the temperature values in order. There are 9 datapoints. You are asked to provide the Gini Index equation if the split is at 16 Celsius.

- a) $2/9 * (1-(2/9)^2-(2/9)^2) + 7/9 * (1-(3/9)^2-(3/9)^2)$
- b) $2/4 * (1-(2/4)^2-(2/4)^2) + 4/5 * (1-(1/5)^2-(4/5)^2)$
- c) $4/9 * (1-(2/4)^2-(2/4)^2) + 5/9 * (1-(4/5)^2-(1/5)^2)$
- d) $4/9 * (1-(2/4)^2-(2/4)^2) + 5/9 * (1-(5/9)^2-(5/9)^2)$

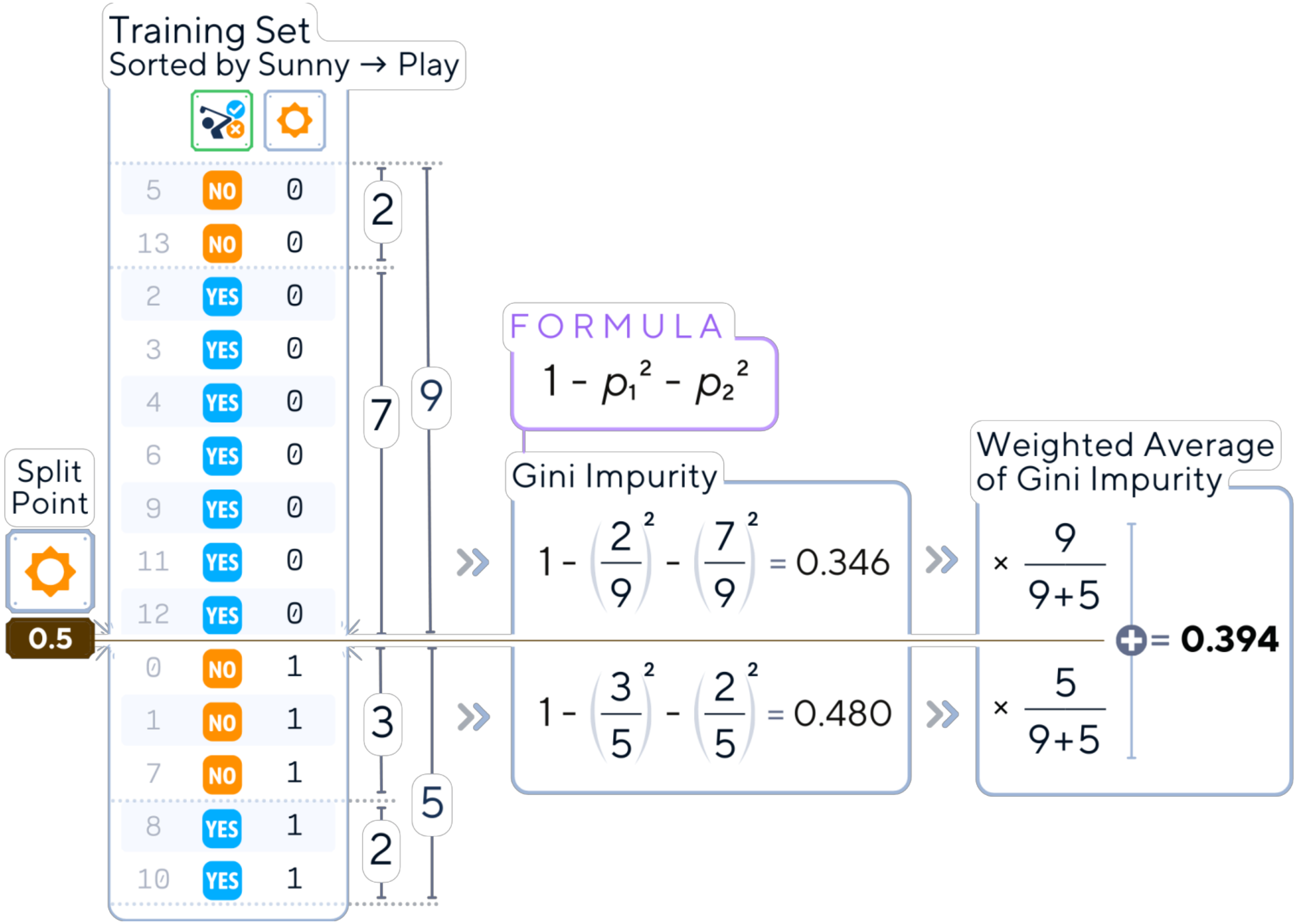
Temp (Celsius)	Play Golf
5	No
10	Yes
12	Yes
15	No
17	Yes
18	Yes
20	Yes
25	Yes
30	No

4

5

3. For each potential split point:

   a. Calculate the impurity (e.g, Gini impurity) of the current node.
   b. Calculate the weighted average of impurities.



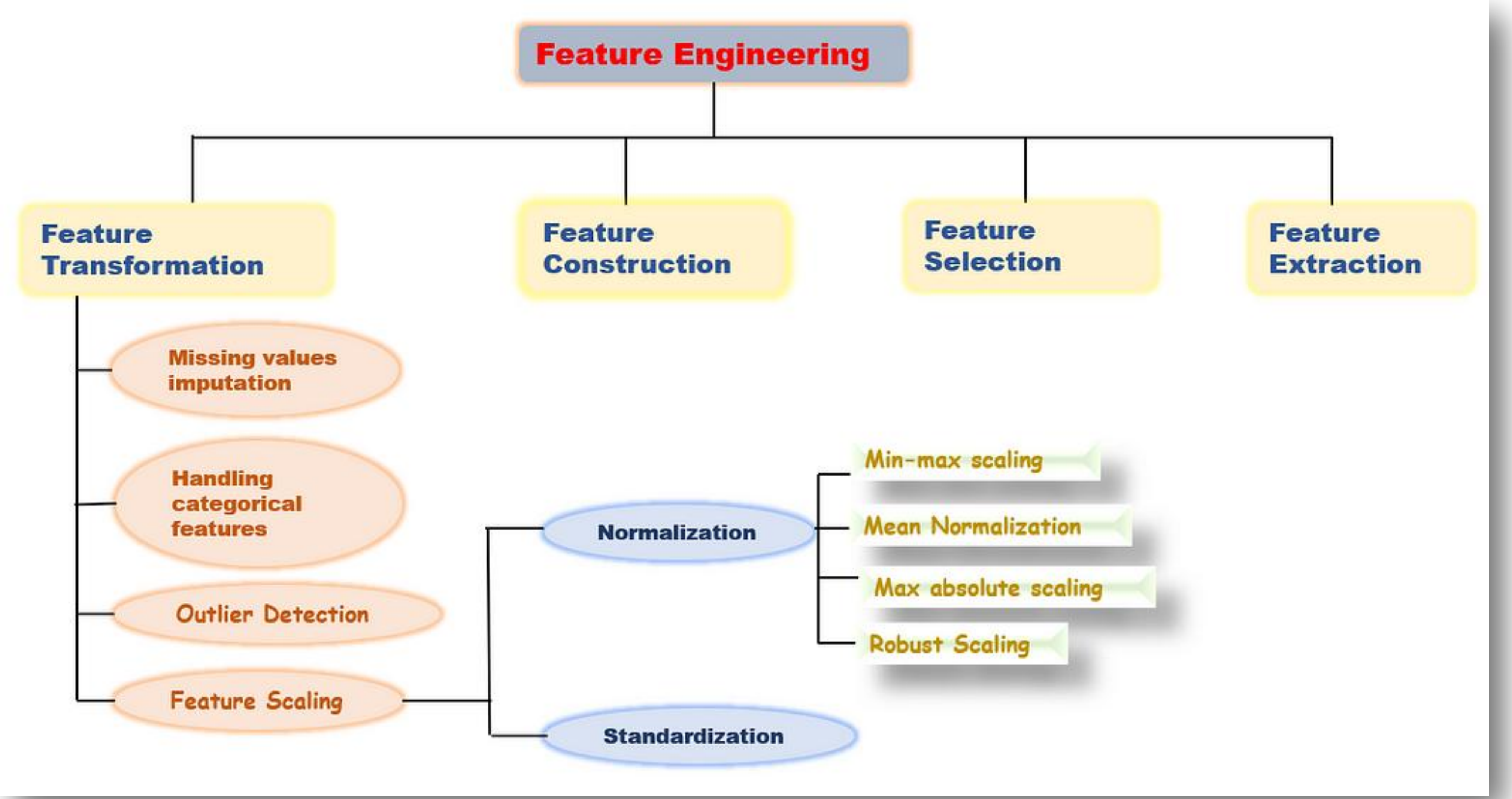
Feature Engineering

Slide deck: Week8-FeatureEngineering

Question 16

Among the 4 types of feature engineering methods, where does changing the scale to a log-scale go?

- a) Feature transformation
- b) Feature construction
- c) Feature selection
- d) Feature Extraction


Feature Engineering
Feature Transformation
Feature Construction
Feature Selection
Feature Extraction
Missing values imputation
Handling categorical features
Outlier Detection
Feature Scaling
Normalization
Standardization
Min-max scaling
Mean Normalization
Max absolute scaling
Robust Scaling

Question 17

Feature normalization limits the range of values for that feature.

- a) True
- b) False

TransactionID	Transaction amount
2199592307	52.32
2273608608	13.99
2325935825	7.49
2557774874	38.57
2678702688	35.01
2881900983	68.12
3038132880	10.21
3123833443	9.99

TransactionID	Normalized transaction amount	Standardized transaction amount
2199592307	0.739	1.008
2273608608	0.107	-0.682
2325935825	0.000	-0.969
2557774874	0.513	0.401
2678702688	0.454	0.245
2881900983	1.000	1.704
3038132880	0.045	-0.849
3123833443	0.041	-0.858

Min Max normalization

Standardization

Ensemble Learning

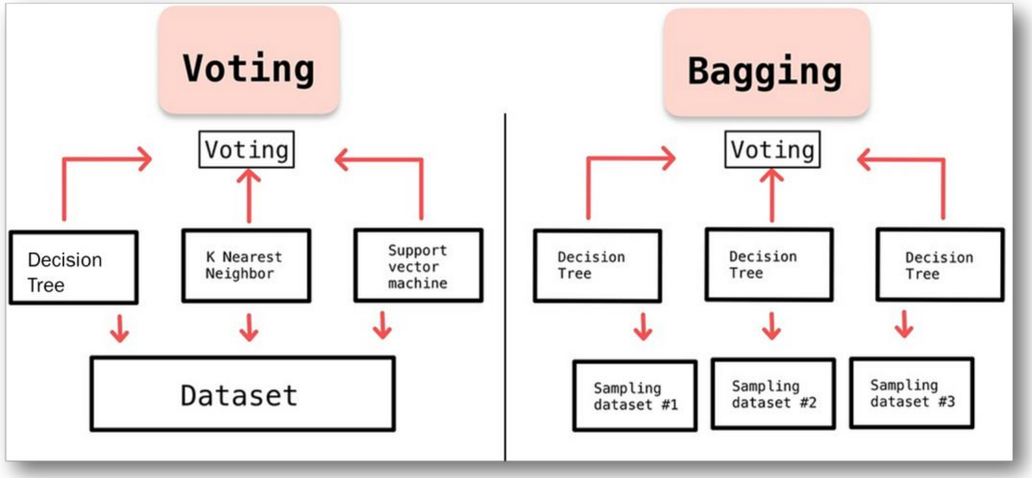
Slide deck: Week8-EnsembleLearning

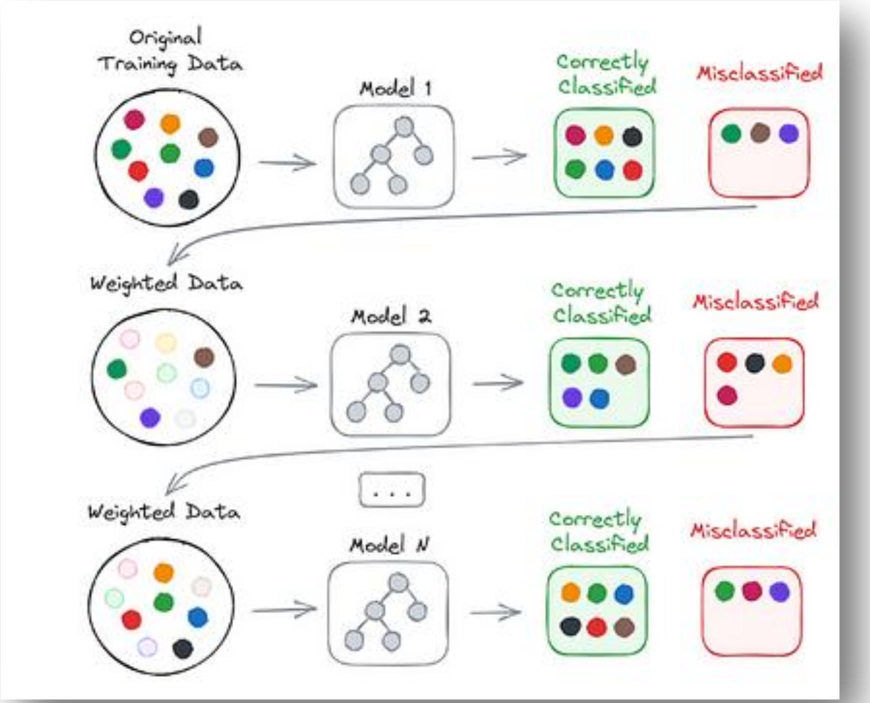
Question 18

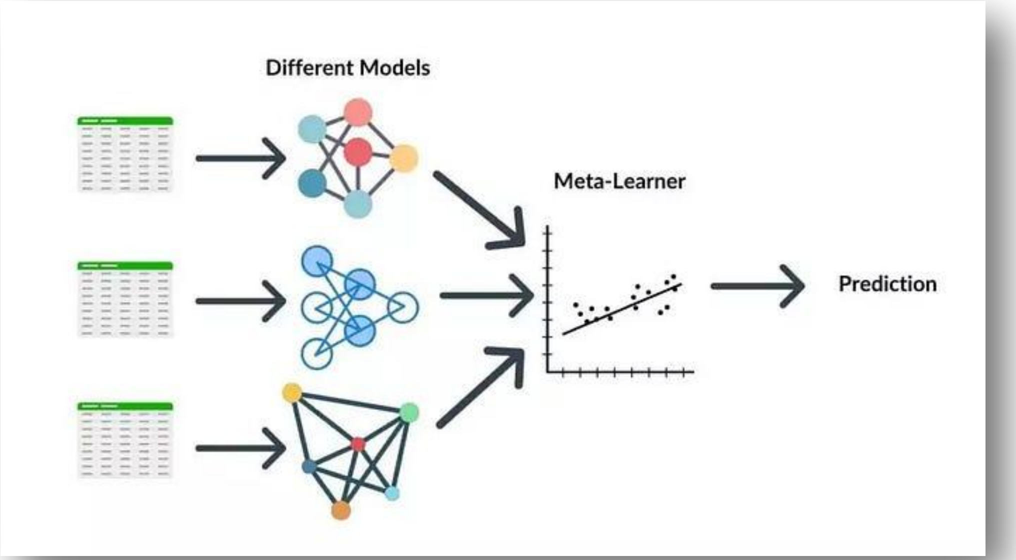
Which of the 4 types of ensemble approach is described or illustrated below?

We iteratively train a decision tree paying more attention, over the iterations, to the misclassified elements.

- a) Voting
- b) Bagging
- c) Stacking
- d) Boosting


Voting
Voting
Decision Tree
K Nearest Neighbor
Support vector machine
Dataset
Bagging
Voting
Decision Tree
Decision Tree
Decision Tree
Sampling dataset #1
Sampling dataset #2
Sampling dataset #3


Original Training Data
Model 1
Correctly Classified
Misclassified
Weighted Data
Model 2
Correctly Classified
Misclassified
...
Weighted Data
Model N
Correctly Classified
Misclassified


Different Models
Meta-Learner
Prediction

Question 19

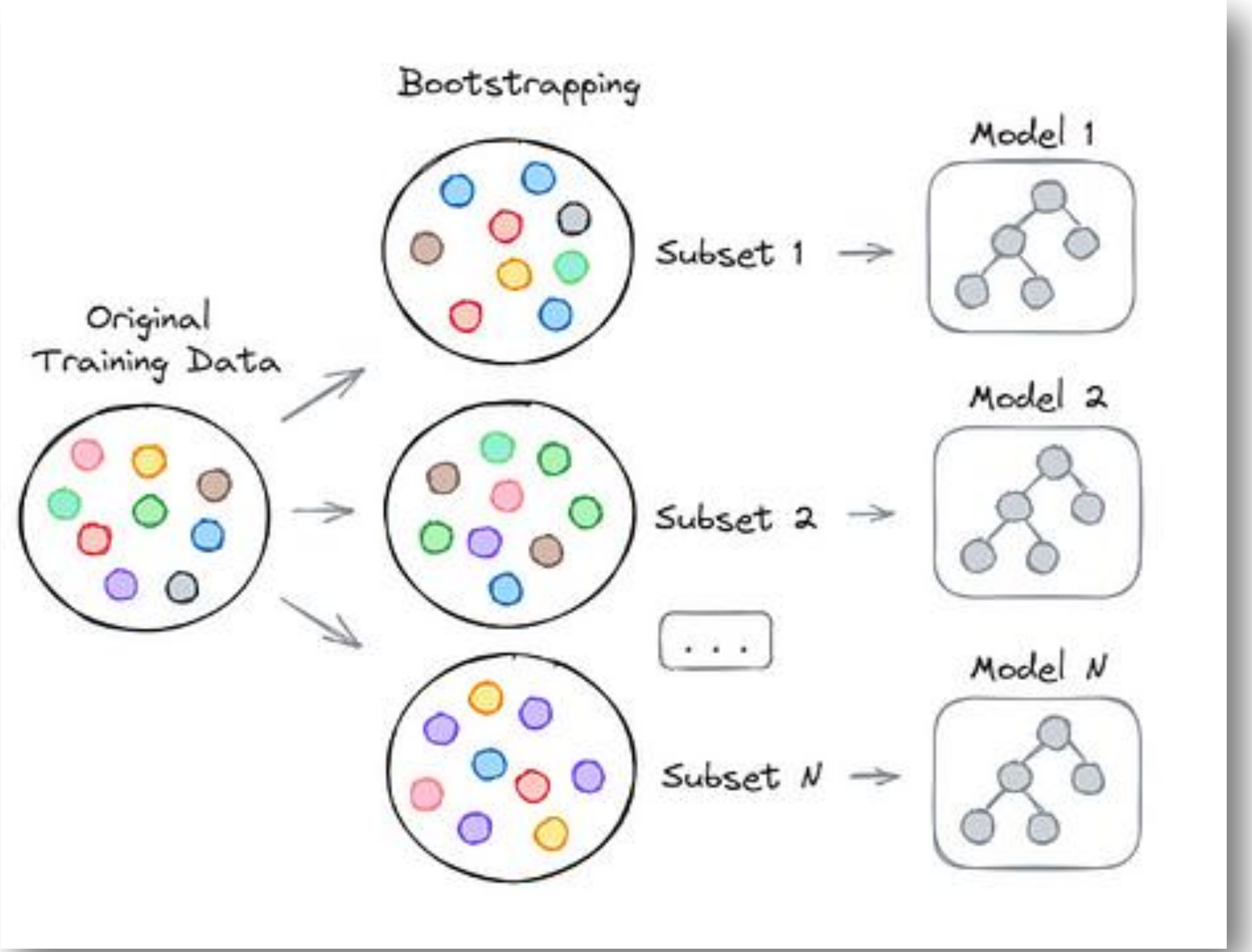
Imagine the following scenario for performing bagging as an ensemble technique.

Imagine a dataset of M elements (X1 to XM), from which N subsets are generated.

Select which statements are true.

- [ ] a) N-1 models will be trained and their outputs will be combined by using the Nth model.
- [ ] b) A datapoint Xi will only be part of one of the N subsets.

Both are false


Bootstrapping
Original Training Data
Subset 1
Subset 2
Subset N
. . .
Model 1
Model 2
Model N

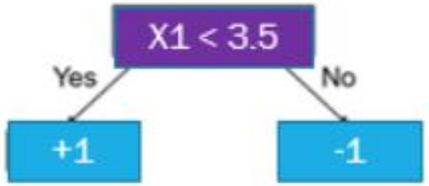
Question 20

We are implementing a boosting algorithm (Adaboost).

Assume the following small dataset, with all datapoints having an equal weight:

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1

Assume in iteration 1, we found a first stump



$0.5 \ln\left(\frac{1-0.2}{0.2}\right)$

Based on this stump, sample 5 is misclassified. What is the importance factor value then, assuming a learning rate of 0.5.

- a) 0.693
- b) 1.098
- c) 0.424
- d) 0.549

Calculate an importance factor

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1

Learning rate

$$\alpha_t = \eta \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

Importance Factor

= 0.5 * ln ((1 – 0.2) / 0.2) = 0.693

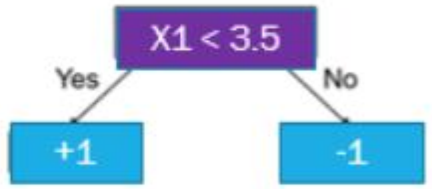
Question 21

We are implementing a boosting algorithm (Adaboost).

Assume the following small dataset, with all datapoints having an equal weight:

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1

Assume in iteration 1, we found a first stump



Based on this stump, assuming an importance factor of 0.3, what is the new weight (before normalization) of the correctly classified data point 4?

- a) 0.148
- b) 0.270
- c) 0.172
- d) 0.862

Readadjust the weights of each data point

$$D_{t+1}(i) = \frac{D_t(i)\exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

Use the importance factor to calculate the new weights

sample	x1	x2	class
1	3	4	1
2	3.2	2	1
3	3.7	1	-1
4	4	4.5	-1
5	4.5	6	1

Current weight = 0.2
Importance factor = 0.3

Correct samples: 0.2 * exp (-0.3) = 0.148
Incorrect samples: 0.2 * exp(+0.3) = 0.270

Z = 4*0.270 + 1 * 0.148 = 1.228

Correct samples: 0.148 / 1.228 = 0.120
Incorrect samples: 0.270 / 1.228 = 0.220

Clustering

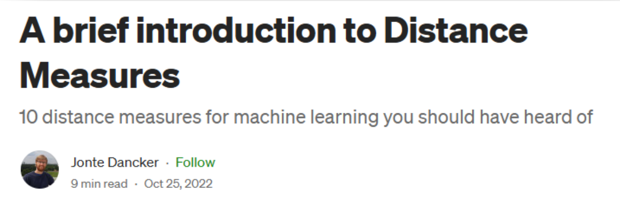
Slide deck: Week9-Clustering

Question 22

Which of the following similarity measures is appropriate for:

Calculating the similarity between DNA sequences?

- a) Euclidian
- b) Manhattan
- c) Hamming
- d) Jaccard



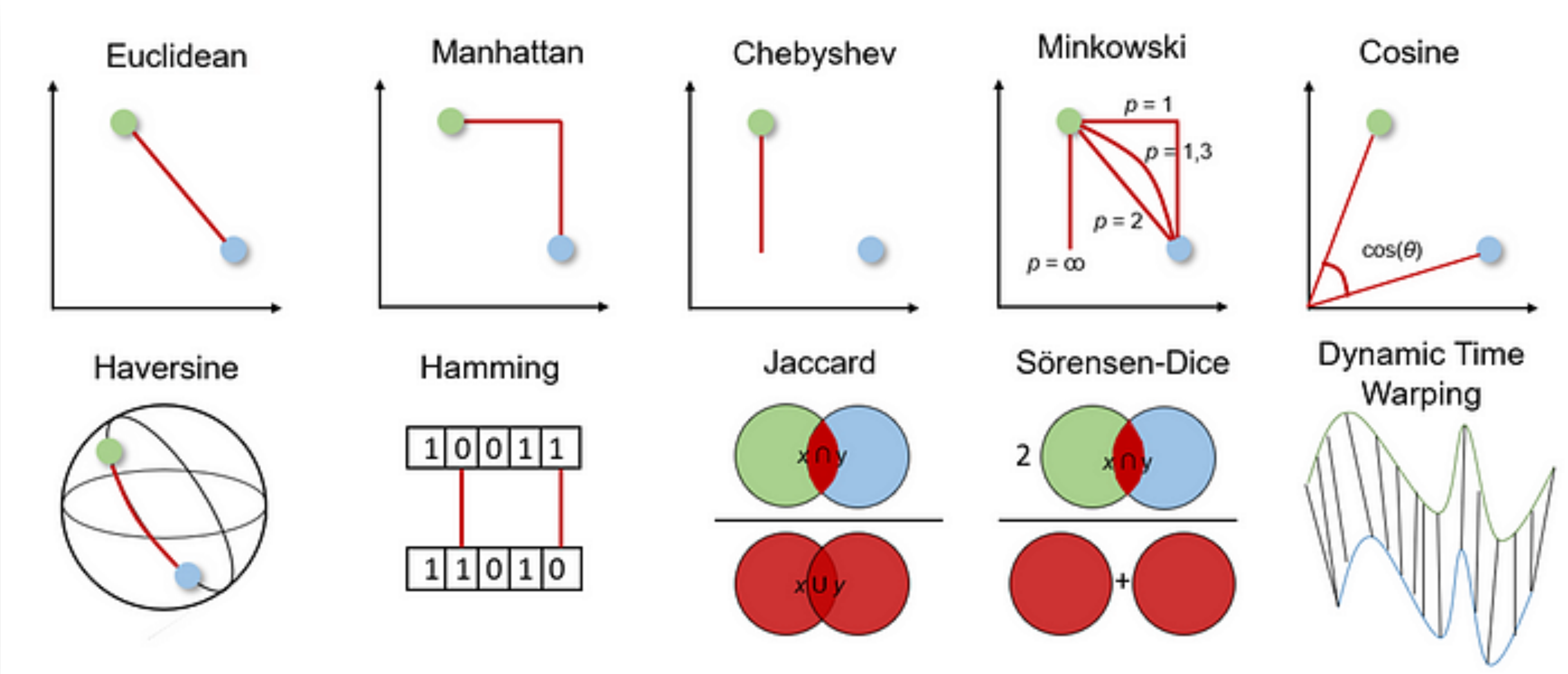
A brief introduction to Distance Measures

10 distance measures for machine learning you should have heard of

Jonte Dancker · Follow
9 min read · Oct 25, 2022

[Source]

Which similarity to use?



Question 23

At the end of kmeans, what would the centroid of each cluster represent?

a) The average distance of all the points to each other in the cluster.

b) The number of points in the cluster.

c) The average position of all the points in the cluster, according to each dimension.

d) The average distance of all the points from that cluster to the other clusters.

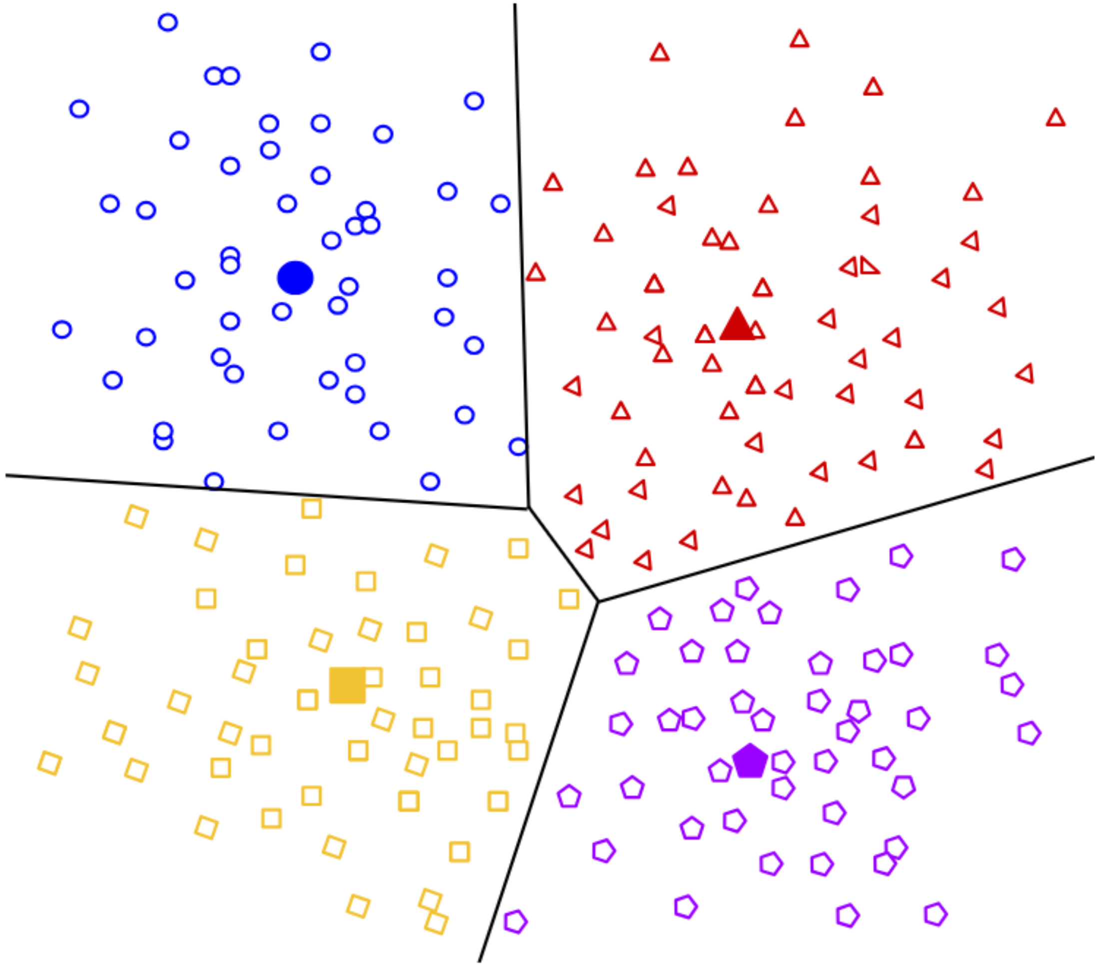


Figure 1: Example of centroid-based clustering.

Example of a result with 4 clusters

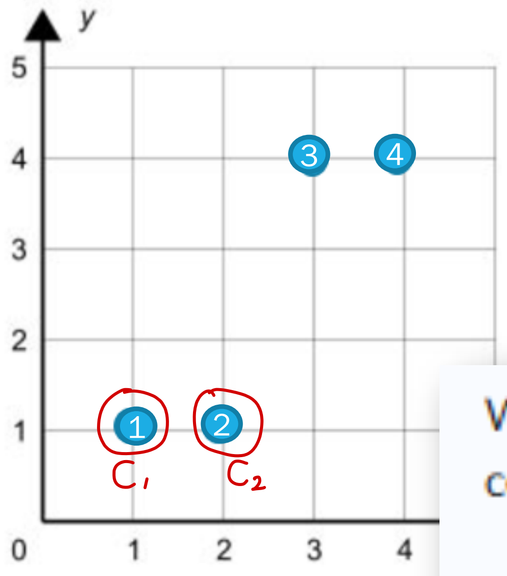
Question 24

Assume the small following dataset on which we will apply the kmeans algorithm:

Sample	X	Y
1	1	1
2	2	1
3	3	4
4	4	4

Assume k=2. Assume the first random selection of centroids is C1=S1 and C2=S2.

Assuming Euclidian Distance.



What will be the positions of the new centroids after the first iteration?

- a) C1 = (1,5), C2 = (5,1)
- b) C1 = (1.5,1) , C2 = (3.5,4)
- c) C1 = (1,1), C2 = (2,1)
- d) C1 = (1,1), C2 = (3,3)

4. For each cluster, calculate a new centroid by taking the mean position of all points in the cluster.

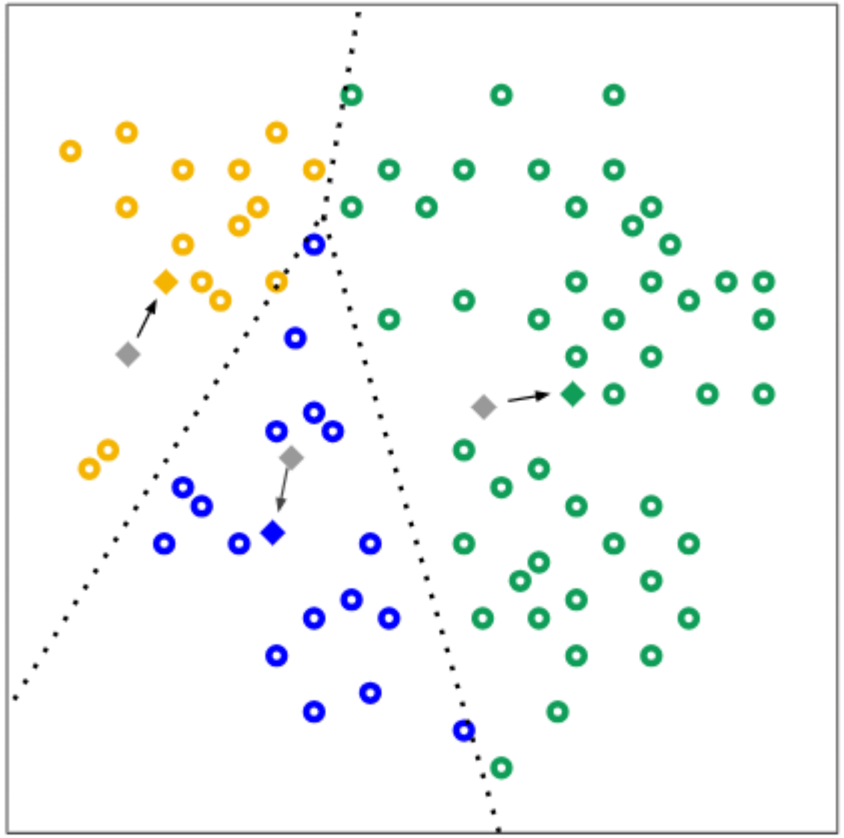
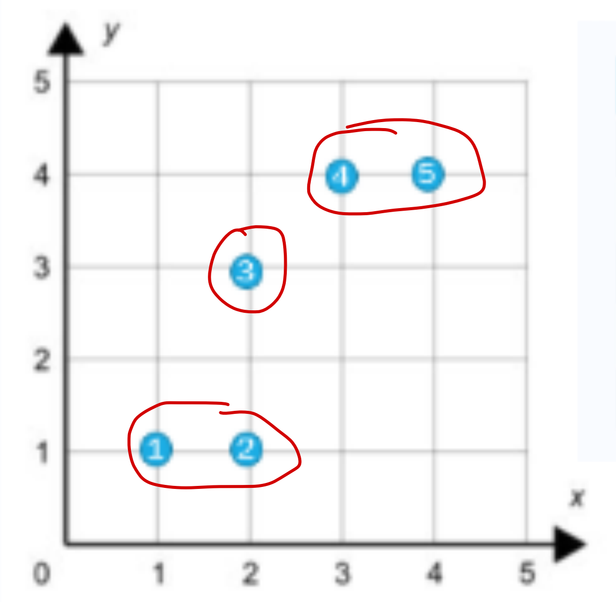


Figure 3: Recomputed centroids.

Question 25

Assume we use Manhattan Distance (MD). an eps of 1 (MD=1) and a min_samples of 2.

We wish to perform DBSCAN clustering on the following dataset:



Sample	X	Y
1	1	1
2	2	1
3	2	3
4	3	4
5	4	4

How would the datapoints be labeled?

- a) Core (1,2,4,5), Border (3), Outlier()
- b) Core (1,2,4,5), Border (), Outlier(3)
- c) Core (1,2,3,4,5), Border (), Outlier()
- d) Core (1,4), Border (2,5), Outlier(3)

The DBSCAN algorithm is built according to the following definitions:

- **Core points**: These are data points inside a cluster. A data point is core if it is surrounded by a minimum number of other data points within a set distance. When you use DBSCAN you set the distance (known as **eps**) and the minimum number of points within the distance (known as **min_samples**).
- **Border points**: A border point is not a core point, but falls within the set distance (i.e., eps) of a core point.
- **Noise points**: A noise point is any point that is neither a core point nor a border point.

Recommender Systems

Slide deck: Week9-Recommendation

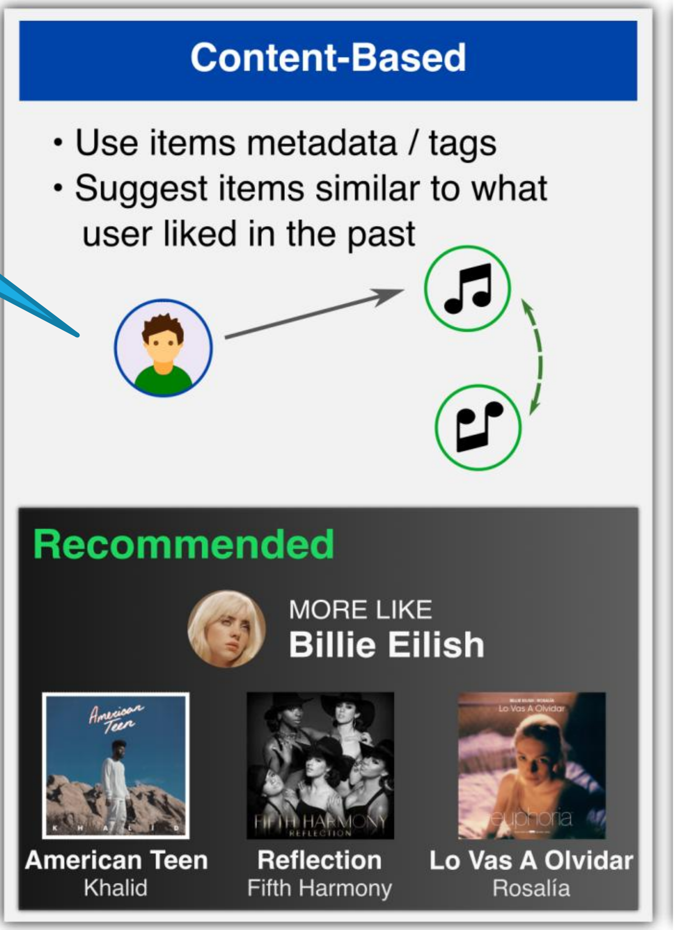
Question 26

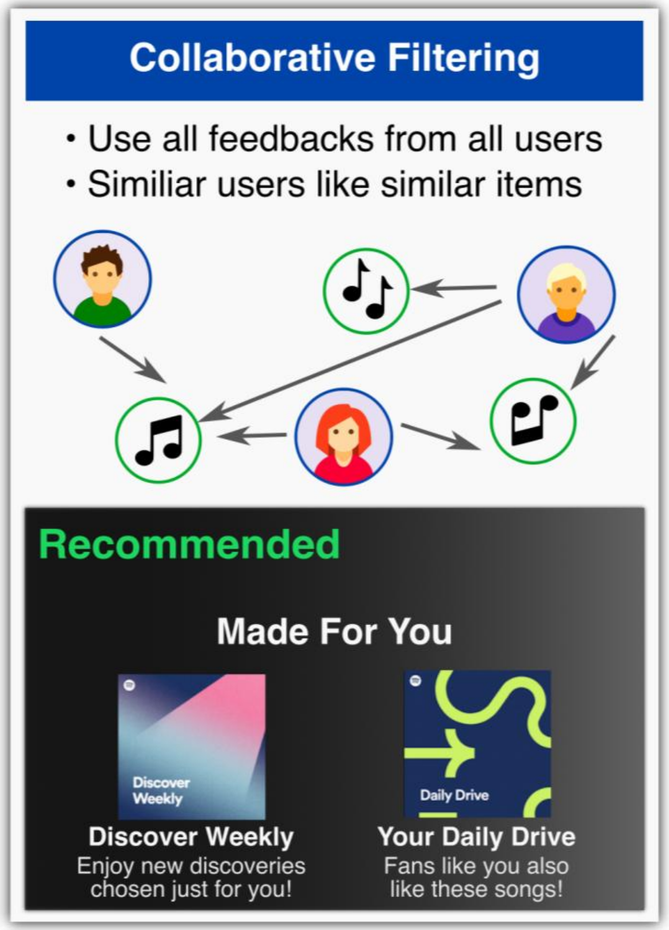
Select what is true about content-based versus collaborative filtering.

a) Content-Based requires item metadata.

b) Content-Based is better in cold start situations.

Personalized recommendation





Hybrid

Question 27

What can you say about the following matrix?

?	?	4	?	1
4	?	?	?	?
?	?	?	3	2
1	?	?	?	?

- a) One of the user only rated one movie.
- b) The raters were asked for explicit feedback.

Explicit Feedback

- Thumbs up / down
- Star ratings from 1 to 5
- Written reviews

Customer Reviews

★★★★☆ 4.0

5 star		62%
4 star		15%
3 star		8%
2 star		1%
1 star		14%

Implicit Feedback

- Played songs / videos
- Purchased items
- Browsing history

Your Browsing History

Sterlizia
Potted Plant

SWATCH #SB02
Man Watch

The Tempest
Book

Question 28

When using this utility matrix to generate an item-item similarity matrix, the resulting matrix will be of size?

?	?	4	?	1
4	?	?	?	?
?	?	?	3	2
1	?	?	?	?

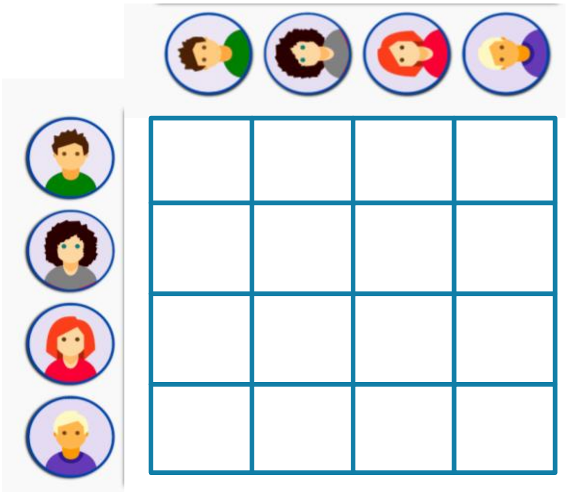
- a) 4x5
- b) 4x4
- c) 5x4
- d) 5x5

Other similarity (e.g. Pearson correlation) can be used

$$similarity(A,B) = \frac{A \cdot B}{\|A\| \times \|B\|} = \frac{\sum_{i=1}^{n} A_i \times B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \times \sqrt{\sum_{i=1}^{n} B_i^2}}$$

	?	?	4	?	1
	4	?	?	?	?
	?	?	?	3	2
	1	?	?	?	?

Item-Item Similarity



User-User Similarity

Question 29

Matrix factorization has been performed:

	M1	M2	M3	M4	M5
F1	3	1	1	3	1
F2	1	2	4	1	3

	F1	F2
A	1	0
B	0	1
C	1	0
D	1	1

	M1	M2	M3	M4	M5
A	3		1		1
B	1	2	4	1	3
C	3	1		3	1
D		3		4	4

Which movie(s) would be recommended to user B assuming 3 is the minimum rating for a recommendation.

- a) M5 only
- b) M2 and M5
- c) M3 only
- d) No movies

	M1	M2	M3	M4	M5
F1	3	1	1	3	1
F2	1	2	4	1	3

What are the recommendations

	F1	F2
A	1	0
B	0	1
C	1	0
D	1	1

	M1	M2	M3	M4	M5
A	3	1	1	3	1
B	1	2	4	1	3
C	3	1	1	3	1
D	4	3	5	4	4

Ranked results

1. M4
2. M2

1. M3
2. M1

Question 30

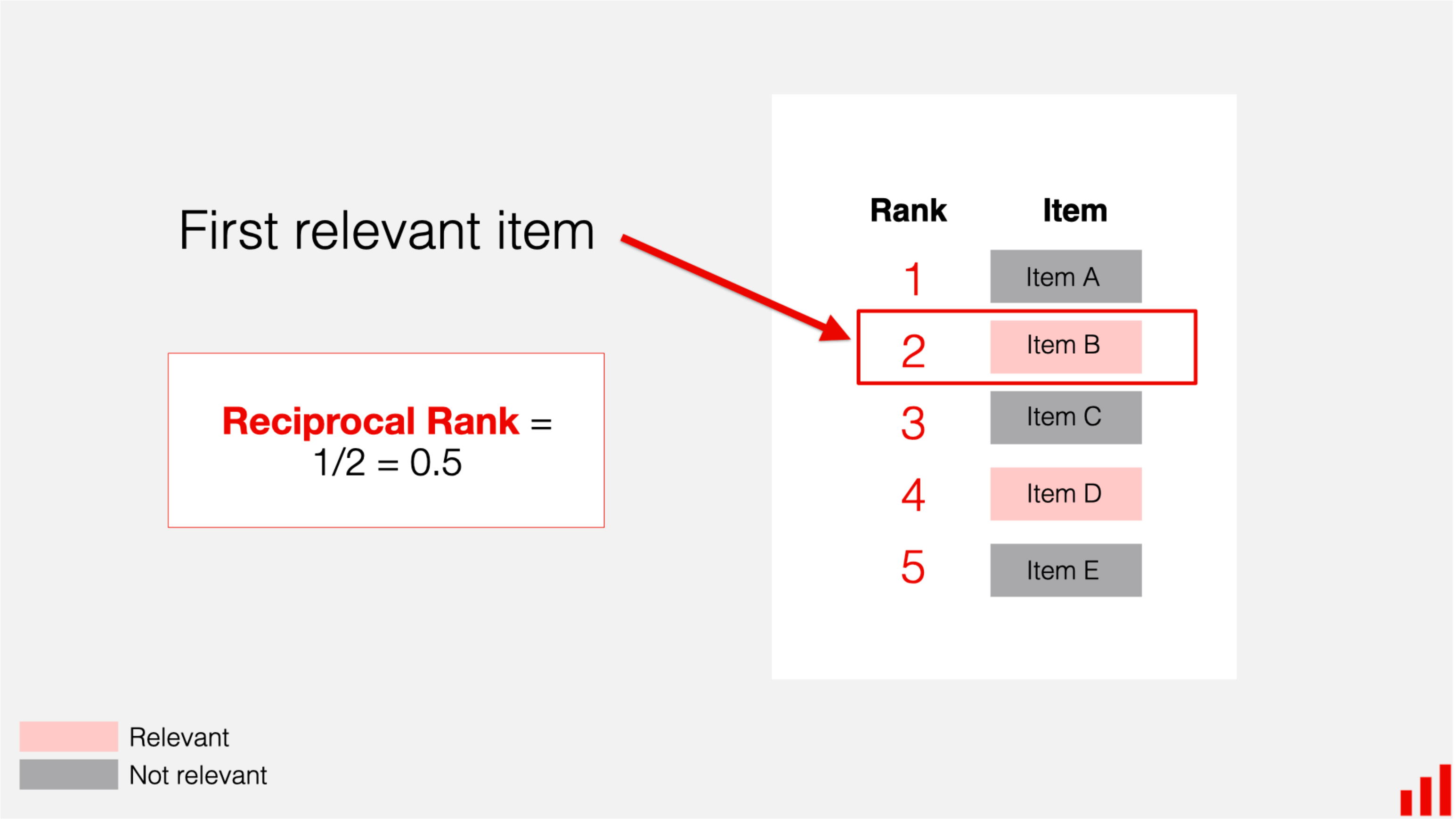
Assume the following results of a recommendation system.

Rank	Movie	Rating
1	MA	2
2	MB	5
3	MC	3
4	MD	4
5	ME	5
6	MF	3
7	MG	2
8	MH	2
9	MI	5
10	MJ	3

Assume relevance is associated with a rating of 3 or more (3, 4 or 5). What is the Reciprocal Rank.

- a) 1/2
- b) 1/3
- c) 1
- d) 1/10

MRR (Mean Reciprocal Rank) shows how soon you can find the first relevant item.
To calculate MRR, you take the reciprocal of the rank of the first relevant item and average this value across all queries or users.



REVIEW SESSION

- Go through 30 example questions for the different weeks
- Point to important material throughout these questions
- Point to question styles throughout these questions